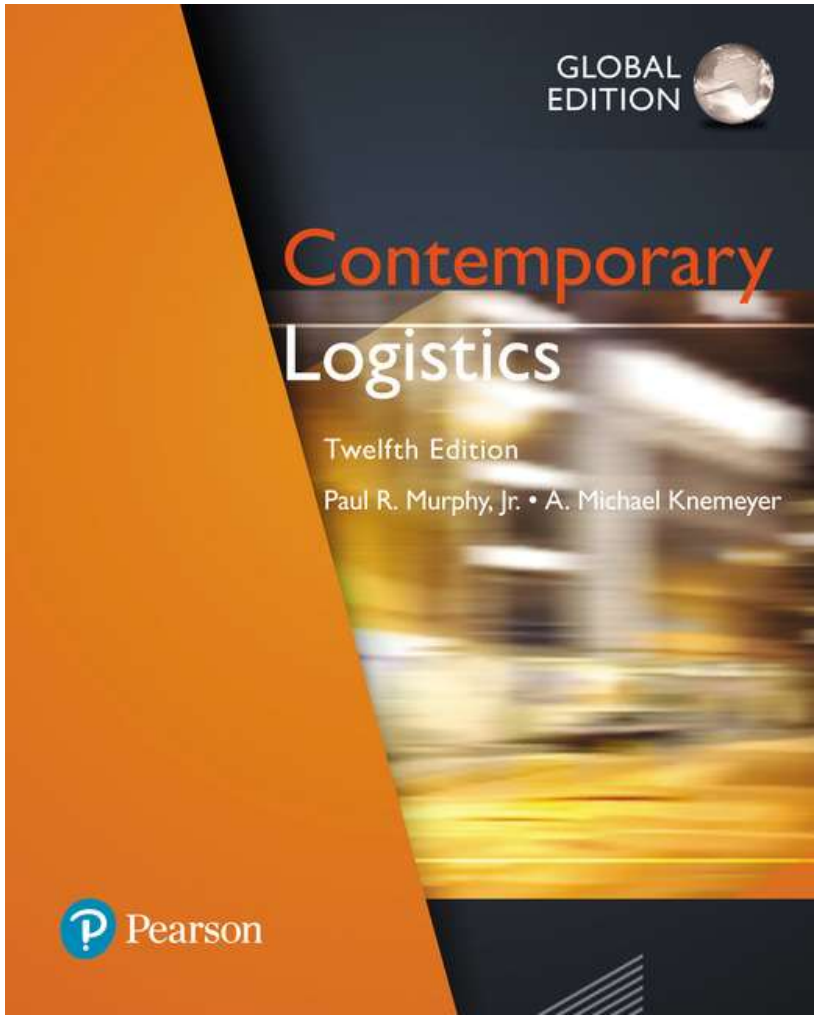
Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 9

Facility Location

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Learning Objectives (1 of 2)

9.1 To examine the strategic importance of facility location

9.2 To explain the general process of determining the optimum number of facilities

9.3 To describe the major factors that influence facility location

9.4 To examine a site's specialized location characteristics

9.5 To explain location decisions using simple grid systems

9.6 To learn about facility relocation and facility closing

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Facility Location

- Facility location
  - Refers to choosing the locations for distribution centers, warehouses, and production facilities to facilitate logistical effectiveness and efficiency
  - Are these strategic or tactical decisions?
- Factors that can influence facility location decisions include:
  - Costs consideration
  - Customer service expectations
  - Location of Customer or Supply Markets

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The Strategic Importance of Facility Location (1 of 3)

- Factors that can influence facility location decisions include:
  - Cost considerations
    - Consumers sensitized to buy only when prices are low
    - Low price/low-cost framework led companies to operate in countries with low-cost and skilled labor
    - In recent years, organizations are reexamining the low-cost labor paradigm (nearsourcing)

Lower production and logistics cost means
more centralization
(few facilities with high utilization)

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The Strategic Importance of Facility Location (2 of 3)

- Factors that can influence facility location decisions include:
  - Customer service expectations
    - Expectations continue to increase over time
    - Customers looking for faster and more reliable order cycles
    - Amazon services orders from more than 120 fulfillment centers located in the United States, Europe, and Asia

Higher customer expectations means
more decentralization
(large network with extra capacity)

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The Strategic Importance of Facility Location (3 of 3)

- Factors that can influence facility location decisions include:
  - Location of customer or supply markets
    - Population shift (Mega Cities)
    - Economic growth (China & India 1/3 of the world)
    - Sustainability (locavore strategy purchasing locally grown or produced foods)

Rank	1970	1990	2014
1	California	California	California
2	New York	New York	Texas
3	Pennsylvania	Texas	Florida
4	Illinois	Pennsylvania	New York
5	Texas	Illinois	Illinois

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Determining the Number of Facilities

- Facilities are generally added when service levels become unacceptable
- Software packages exist to help in determining the number and location of facilities
- Key issue—time it takes to get from facility to majority of the U.S. (or target) population
  - Moving from 1 to 3 facilities saves 1 day lead time
  - Moving from 3 to 10 facilities saves ¼ day lead time[1]

[1]www.chicago-consulting.com

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Figure 9.1: Chicago Consulting's 10 Best Warehouse Networks

THE 10 BEST CHINESE WAREHOUSE NETWORKS
Networks with the Lowest Possible "Time-to-the-Chinese Population"

Number of Warehouses	Average Distance to the Chinese Population (Miles)	Average Transit Lead-Time to the Chinese Population (Days)	Best Warehouse Locations		
ONE	504	3.38	XINYANG		
TWO	377	2.55	LIANYUAN	FEICHENG	
THREE	309	2.15	PINGXIANG	JINAN	ZIYANG
FOUR	265	1.87	PINGXIANG CHANGCHUN	JINING	ZIYANG
FIVE	228	1.65	SHAOGUAN CHANGCHUN	HANDAN NANJING	ZIYANG
SIX	207	1.53	SHAOGUAN CHANGCHUN	HANDAN NANJING	NEIJIANG URUMQI
SEVEN	184	1.42	GUANGZHOU CHANGCHUN HONGHU	HANDAN JINGJIANG	NEIJIANG URUMQI
EIGHT	168	1.31	GUANGZHOU CHANGCHUN HONGHU	LIAOCHENG YIXING BAOJI	YIBIN URUMQI
NINE	154	1.24	BEILIU CHANGCHUN YUEYANG	LIAOCHENG LIYANG BAOJI	YIBIN URUMQI ZHANGZHOU
TEN	141	1.20	BEILIU CHANGCHUN YUEYANG TIANJIN	KAIFENG YIXING BAOJI	YIBIN URUMQI ZHANGZHOU

The 10 Best Chinese Warehouse Networks have been developed based on the lowest possible transit lead-times to "customers" represented by the Chinese population. For example, Xinyang provides the lowest possible lead-time for one warehouse. Any other place will increase transit lead-time to the Chinese population. Similarly putting any three warehouses in any locations other than Pingxiang, Jinan or Ziyang will cause the transit lead-time to be higher than 2.15 days.

© Chicago Consulting
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[illegible]

Chicago Consulting

Source: Courtesy of Terry Harris, Managing Partner of Chicago Consulting.

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General Factors Influencing Facility Location (1 of 7)

- Raw materials, Component parts, Labor, Markets influence on location of manufacturing, processing, or assembly facility
- Location of warehouses, distribution centers, and cross-docking facilities are influenced by locations of plants whose products they handle and markets they serve
- Taxes and incentives, Inventory tax, Incentive packages
- Transportation availability refers to the number of transportation modes (intermodal competition) and the number of carriers within each mode (intramodal competition) and their cost
- Proximity to industry clusters
- Trade patterns and trade agreements
- Quality-of-life considerations (business ecosystem)
- Locating in other countries: Relevant legal system, Political stability, Bureaucratic red tape, Corruption, Protectionism, Nationalism, Privatization, confiscation, Treaties and trade agreements

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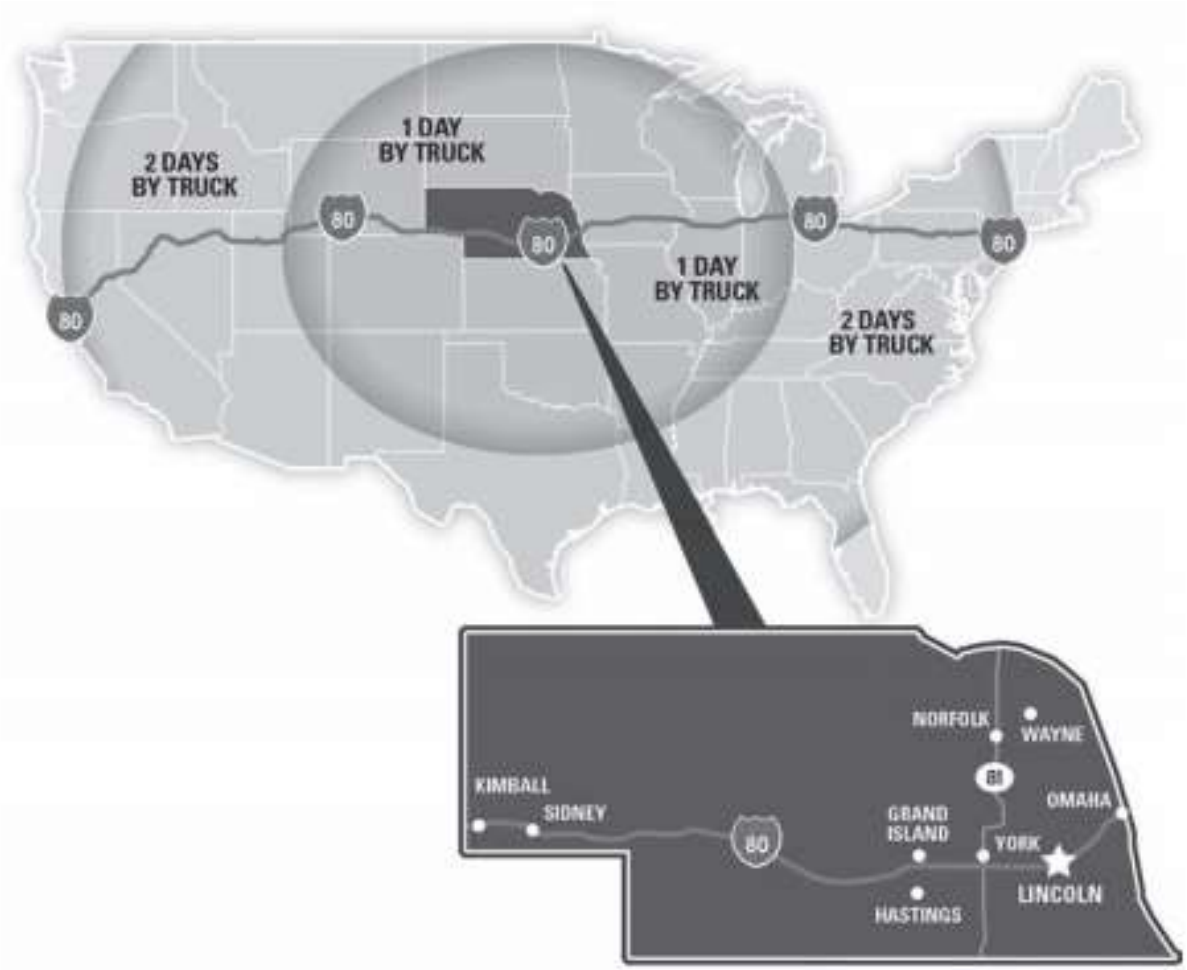
Figure 9.2: Truck Distances from Nebraska



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Specialized Location Characteristics (2 of 4)

- Title search
  - May be needed to make sure a parcel of land can be sold and that there are no liens against it
- Engineers examine site for proper drainage and to determine load-bearing characteristics of the soil
- Due diligence of environmental factors
  - Brownfields
    - Previously used sites that contain chemicals or other types of industrial waste

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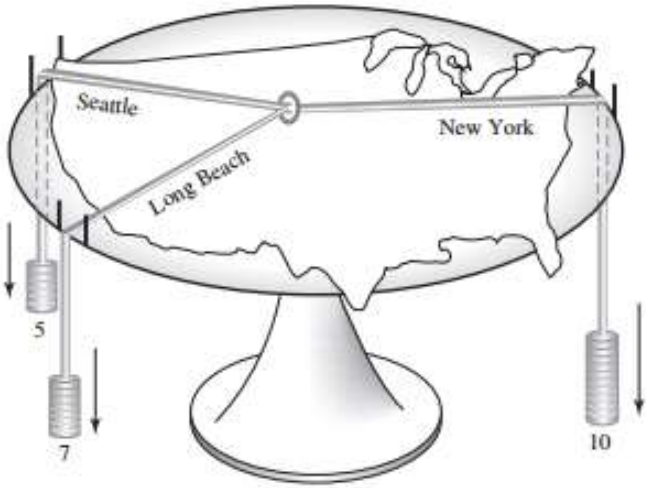
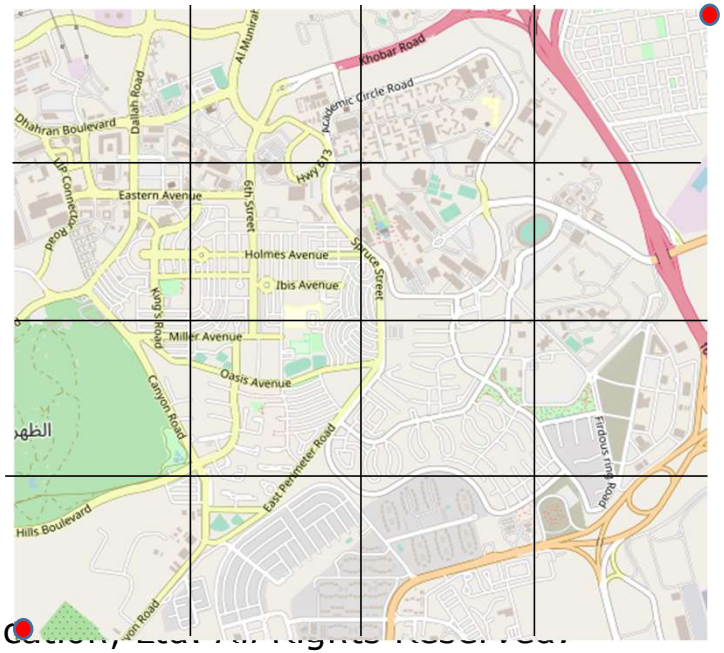
Finding the Lowest-Cost Location Using Grid Systems

- Center-of-Gravity location for a Warehouse (Factory) Serving Retail Stores (Market)
- Grid systems
  - Allow one to analyze spatial relationships with mathematical tools
  - Checkerboard patterns
  - Placed so that they coincide with north–south and east–west line on a map



(50.1594, 26.3181)



(50.1351, 26.2912)

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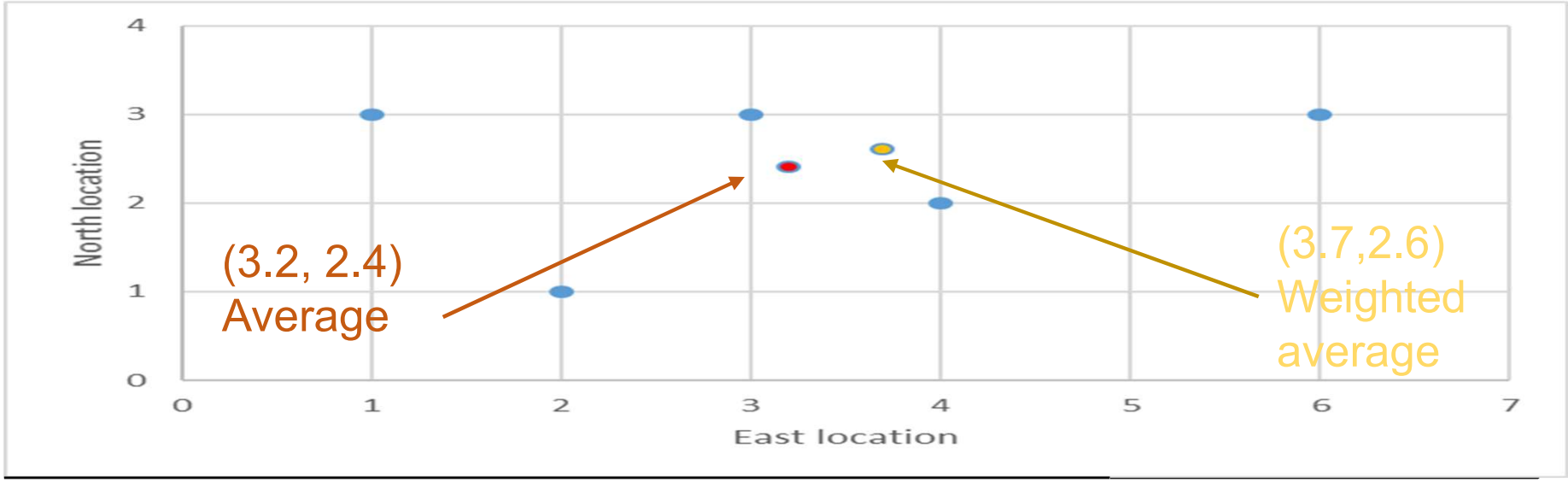
Example: Center-of-Gravity Location for a Warehouse Serving Five Retail Stores



Store	East location	North location	Monthly volume	East x Volume	North x Volume
1	1	3	3	3	9
2	3	3	5	15	15
3	4	2	4	16	8
4	2	1	2	4	2
5	6	3	6	36	18
Total			20	74	52
	3.2	2.4		3.7	2.6
	Average			Weighted average	

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Location Problems

- Single facility location problem:
- Given **m existing facilities**, find the best location of **one new facility**. The best location will be determined based on some distance criteria.
- Solution space
  - Continuous vs Discrete.
- Criteria
  - **Minisum criteria**: the weighted sum of distances among pairs of facilities is minimized.
  - **Minimax criteria**: the weighted maximum distances traveled from new facilities to the existing facilities or to the new facilities is minimized.
  - **Maximin criteria**: the weighted minimum distances traveled from new facilities to the existing facilities or to the new facilities is maximized.

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Location Problems

- Distance measures
  - Rectilinear distance measure :

Distances are measured along paths that are perpendicular to one another. Let (x, y) and (a, b) be two points, then the distance measure is

$d = |x – a| + |y – b|$

  - Euclidean distance measure:

Distances are measured along straight lines. Let (x, y) and (a, b) be two points, then the distance is defined as

$d = [(x - a)^2 + (y - b)^2]^{1/2}$

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Location Problems

- Other distance measures:
  - Squared Euclidean distance measure:
    - $d = [(x - a)^2 + (y - b)^2]$
  - Chebyshev distance measure:
    - $d = \max(|x – a| ; |y – b|)$
  - Actual distance measure:
    - Route distance on Google maps.

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Location Problems

- **Idea**: Given *m* existing facilities, place one new facility
- such that it optimizes a specific criterion.
- **Notations:**
- **$X = (x, y)$** : location of new facility
- **$P_i = (a_i , b_i)$** : location of existing facility *i* , *i* =1, .. ,*m*
- **w_i** : weight associated with travel between new facility and existing facility *i*
- **$d(X, P_i)$** : distance between new facility and existing facility i
- **$f (x, y)$** : the objective function to be minimized

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Location Problems – Rectilinear Minisum

- **Rectilinear Minisum**

$$\text{minimize}: \sum_{i=1}^{m} w_i[|x - a_i| + |y - b_i|]$$

$$\text{subject to}: \; x, \; y \in \mathbb{R}$$

- **The goal is to minimize the total weighted rectilinear distance.**
- **How to solve it?**
  - Median Rule
  - Cumulative weight diagram
  - Resultant force diagram

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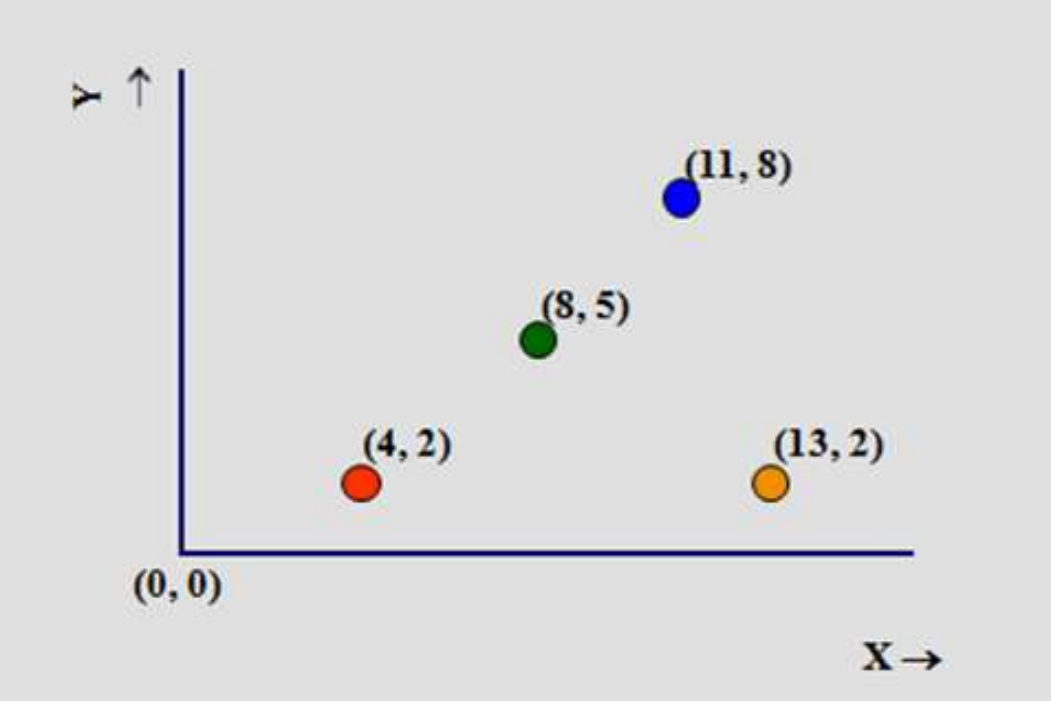
Location Problem: Rectilinear Minisum Example

- Supermarkets 1,2,3, and 4 are located at the points (8,5), (4,2), (11,8), and (13,2) respectively. There are 9, 6, 4, and 12 trips per week between the supermarkets and the new warehouse. Find the best location for the new warehouse based on the rectilinear distance measure and the minisum criteria.



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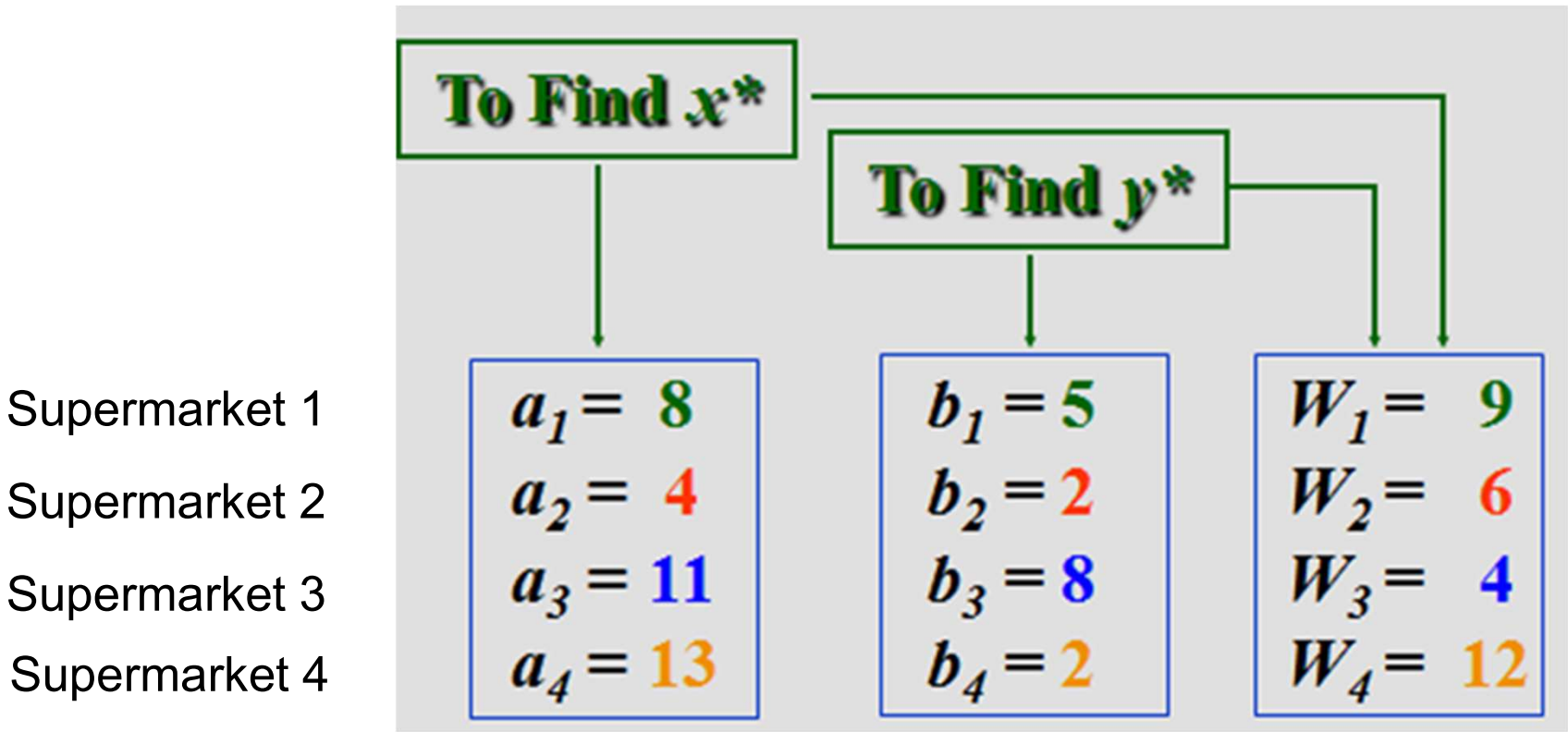
Location Problem: Rectilinear Minisum Example



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Location Problem: Rectilinear Minisum Example

Calculation of Optimal x^*

Supermarket i	Coordinate a_i	Trips w_i	Sum of w_i
2	4	6	6
1	8	9	15
3	11	4	19
4	13	12	31

x^*

$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

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Location Problem: Rectilinear Minisum Example

Calculation of optimal y^*

$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

Supermarket i	Coordinate b_i	Weight w_i	Sum of w_i
2	2	6	6
4	2	12	18
1	5	9	27
3	8	4	31

y^*

Optimal Location: $(x^*, y^*) = (11, 2)$

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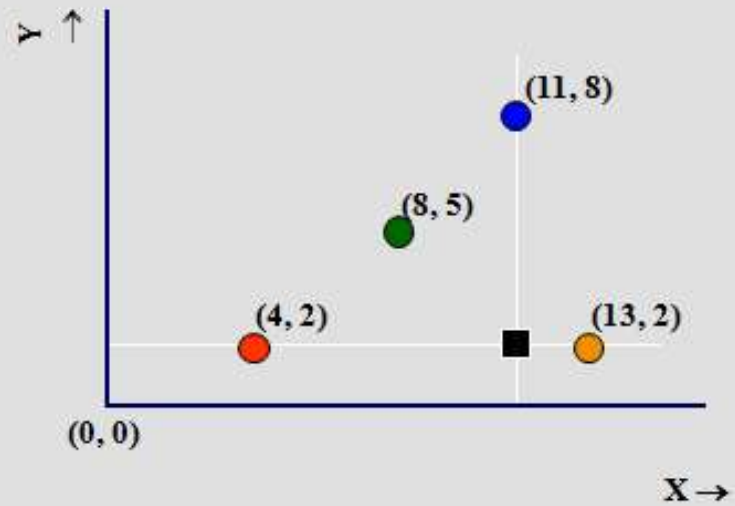
Location Problem: Rectilinear Minisum Example



$$f(x,y) = 9\,|\,11-8\,| + 6\,|\,11-4\,| + 4\,|\,11-11\,| + 12\,|\,11-13\,| +$$
$$9\,|\,2-5\,| + 6\,|\,2-2\,| + 4\,|\,2-8\,| + 12\,|\,2-2\,|$$

$$= 93 + 51$$

$$= 144$$

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Location Problem: Square Euclidean Minisum

- **Square Euclidean Minisum:**

$$\text{minimize}: \sum_{i=1}^{m} w_i[(x-a_i)^2 + (y-b_i)^2]$$

$$\text{subject to}: \; x, \; y \in \mathbb{R}$$

- **The goal is to minimize the total weighted square euclidean distance.**
- **How to solve it?**
  - **Find the min value (set derivative equal to zero and solve)**

$$x^o = \frac{\sum_{i=1}^{m} w_i a_i}{\sum_{i=1}^{m} w_i}$$

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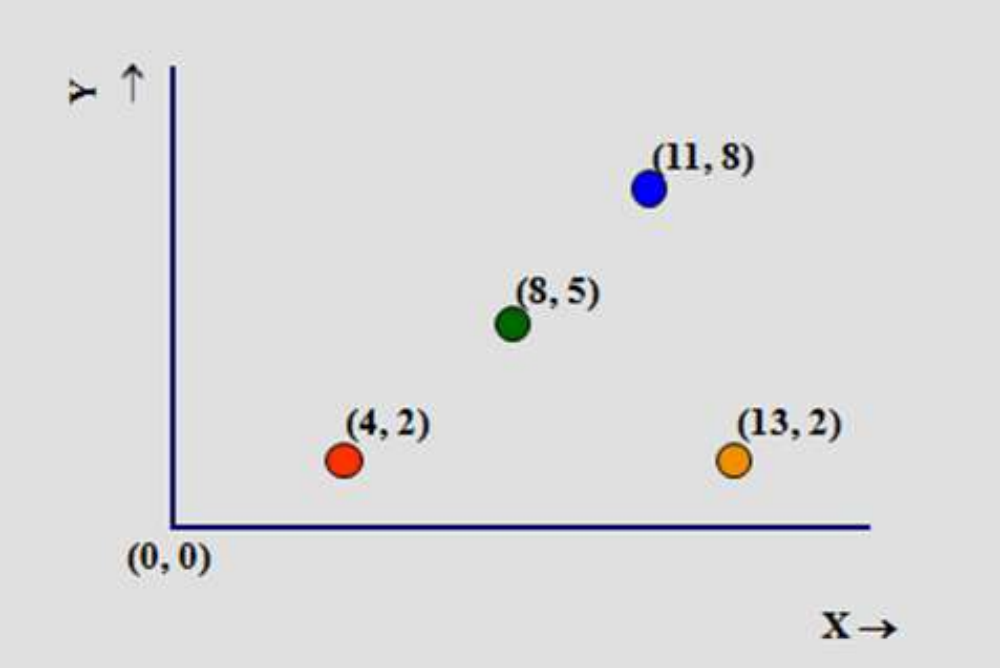
Location Problem: Square Euclidean Minisum Example

- Back to the same example
- Supermarkets 1,2,3, and 4 are located at the points (8,5), (4,2), (11,8), and (13,2) respectively. There are 9, 6, 4, and 12 trips per week between the supermarkets and the new warehouse. Find the best location for the new warehouse based on the square euclidean distance measure and the minisum criteria.



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Location Problem: Square Euclidean Minisum Example

- let $(a_1; b_1) = (8, 5)$, $(a_2, b_2) = (4, 2)$, $(a_3, b_3) = (11, 8)$
- and $(a_4, b_4) = (13, 2)$.
- $w_1 = 9$,$w_2 = 6$,$w_3 = 4$ and $w_4 = 12$.

- $x^* \quad = {\sum_{i=1}^{4} w_i a_i}/{\sum_{i=1}^{4} w_i} = \frac{9*8+6*4+4*11+12*13}{31} = 9.548$
- $y^* \quad = {\sum_{i=1}^{4} w_i b_i}/{\sum_{i=1}^{4} w_i} = \frac{9*5+6*2+4*8+12*2}{31} = 3.645$
- The optimal location w.r.t square euclidean minisum criterion is (x, y) = (9.548, 3.645)

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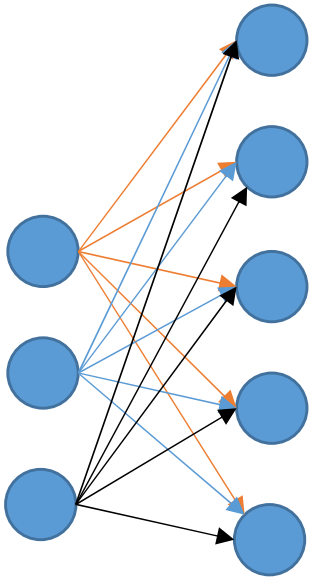
Plants & Markets Allocation Network Optimization



- **Manufacturer of petrochemical with worldwide network in five continents with demand in million liters**
- **Logistics manager has three plants in South America, Asia, and Africa with available capacity (mln liters) to serve demand regions**
- **Cost (inventory, production, transportation) is divided to:**
  1. **Variable cost per mln of liters reported in 1,000$**
  2. **Fixed annual cost (in 1,000$) and capacity**

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Capacity	**Fixed Cost ($)**
S. America	**117**	**77**	**108**	**98**	**100**	20	**6,750**
Asia	**115**	**125**	**90**	**59**	**74**	20	**6,150**
Africa	**142**	**100**	**103**	**105**	**71**	20	**6,000**
Demand (million liters)	12	8	14	16	7		

Mathematical Programming Optimization: Linear Programming (1)

- A furniture dealer deals in only two items–tables and chairs
- Dealer has SR 50,000 to invest and space of at most 60 pieces.
- A table costs SR 2500 and a chair SR 500
- Dealer estimates that the sale of one table makes a profit of SR 250 and of selling one chair SR 75 profit
- Dealer wants to know how many tables & chairs to buy from the available budget to maximize the profit, assuming all items are sold
- Problem formulation
  - Decision variables: x number of tables to buy, y number of chairs to buy
  - Objective function: Maximize $250x + 75y$ (total profit)
  - Constraints: $2500x + 500y \leq 50{,}000$ (Budget constraint)

    $x + y \leq 60$ (storage capacity)
- Problem Type: Linear objective function and constraints in decision variables. Decision variables are continuous. This problem in a linear programming problem
- Solution algorithm: Simplex method implemented numerical software Excel, Lingo, Cplex, Gurobi, Ample, GAMS

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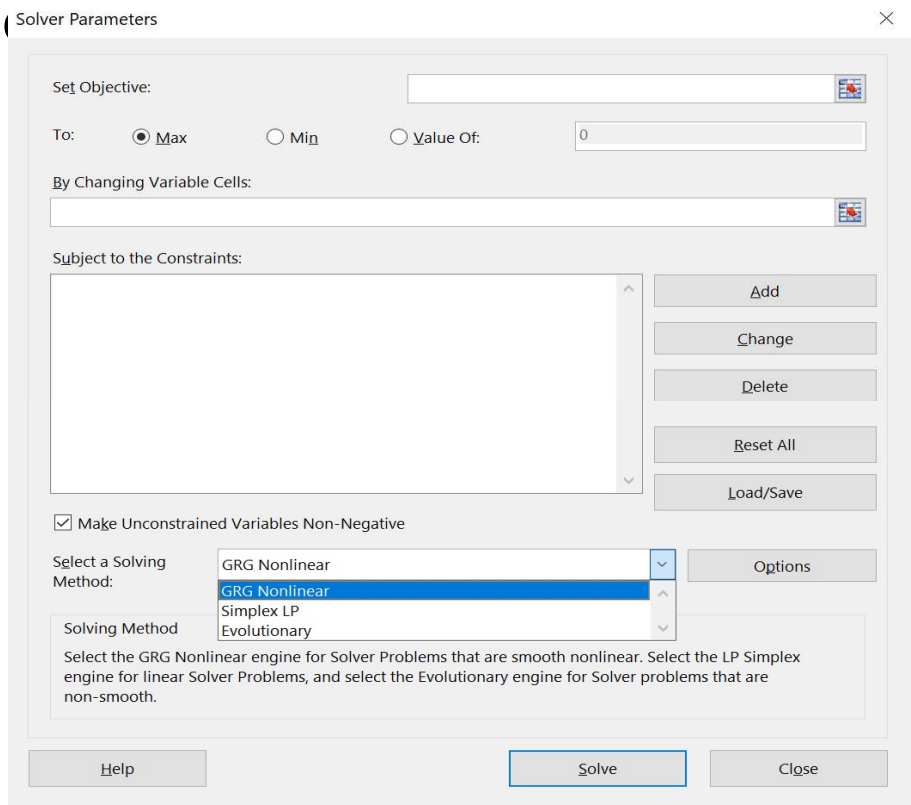
Mathematical Programming Optimization: Linear Programming (2)

- Maximize $250x + 75y$ (total profit)

$$2500x + 500y \leq 50{,}000 \quad \text{(Budget constraint)}$$
$$x + y \leq 60 \text{ (storage capacity)}$$

- Enable Excel Solver Add-in. You find 'Solver' under 'Data'

1. Designate specific Cells for decision variables, objective function, and each of constraints left-hand-sid
2. Configure the 'solver' by setting
3. objective function
4. Selecting types of optimization
5. Setting the decision variables
6. Add the constraint one by one
7. Selecting the algorithm
8. Click Solve



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Plants & Markets Allocation Net. Opt.

- **Decision variables**
  - How much to produce and transport from a plant to a region?
- **Objective**: minimize total costs
- **Constraints**:
  - All demands should be satisfied
  - All plant capacity limits are met

- **Decision variables** x_{ij}
- **Objective**
  Sum $c_{ij}x_{ij}$ for all i, j
- **Constraints**:
  - Unmet demands is 0
  - Left capacity is +

Optimizing the problem using Excel Solver gives

	Demand Region					
Supply Plants	N. America	S. America	Europe	Asia	Africa	Left capacity
S. America	x11 12	x12 8	x13 0	x14 0	x15 0	20 - sum(x11 to x15)>=0
Asia	x21 0	x22 0	x23 4	x24 16	x25 0	20 - sum(x21 to x25)>=0
Africa	x31 0	x32 0	x33 10	x34 0	x35 7	20 - sum(x31 to x35)>=0
Unmet demand	12 - sum(x11 to x31) =0	8 - sum(x12 to x32) =0	14 - sum(x13 to x32) =0	16 - sum(x14 to x34)=0	7 - sum(x15 to x35)=0	

Objective: Sum (c11*x11 to c35*x35) $4,851

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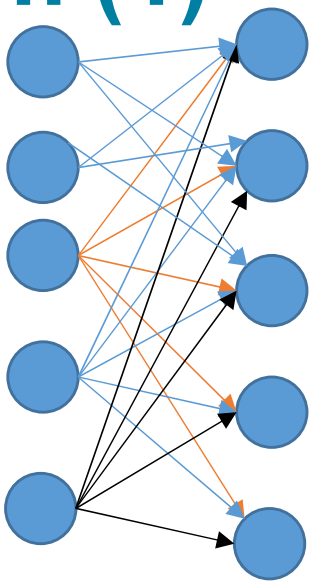
Capacitated Plant Location Problem (1)



- **Manufacturer of petrochemical with worldwide network in five continents with demand in million liters**
- **Logistics manager has <u>five potential</u> plants in North America, South America, Europe, Asia, and Africa with available capacity (mln liters) to serve demand regions**
- **Cost (inventory, production, transportation) is divided to:**
  1. **Variable cost per mln of liters reported in 1,000$**
  2. **Fixed annual cost (in 1,000$) and capacity**

Which plants to open and how to allocate the market demand

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Capacity	**Fixed Cost ($)**
N. America	**81**	**92**	**101**	**130**	**115**	20	**9,000**
S. America	**117**	**77**	**108**	**98**	**100**	20	**6,750**
Europe	**102**	**105**	**95**	**119**	**111**	20	**9,750**
Asia	**115**	**125**	**90**	**59**	**74**	20	**6,150**
Africa	**142**	**100**	**103**	**105**	**71**	20	**6,000**
Demand mln lt	12	8	14	16	7		

Capacitated Plant Location Problem (2)

- **Decision variables**
  - Which plants to open?
  - How much to produce and transport from a plant to a region?
- **Decision variables**
  - z_i: 1 if plant i is open, otherwise.
  - x_{ij}: quantity to ship from i to j
- **Objective**: minimize total costs including variable and fixed costs. <u>Fixed cost is only paid</u> if plant i is open
- **Objective** Sum $f_i z_i$ for all i + Sum $c_{ij} x_{ij}$ for all i, j

  Fixed cost Variable cost
- **Constraints**:
  - All demands should be satisfied
  - All plant capacity limits are met
- **Constraints**:
  - $Unmet\ demand\ is\ 0 \quad (D_j - \sum_{i=1}^{5} x_{ij} = 0)$
  - Left capacity is + $(20z_i - \sum_{j=1}^{5} x_{ij} \geq 0)$

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Capacitated Plant Location Problem (3)

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Left Capacity	**Open / close**
N. America	**X11 12**	**X12 8**	**X13 0**	**X14 0**	**X15 0**	**20z1-sum(x11 to x15)**	**Z1 1**
S. America	**X21 0**	**X22 0**	**X23 0**	**X24 0**	**X25 0**	20z2-sum(x21 to x25)	**Z2 0**
Europe	**X31 0**	**X32 0**	**X33 0**	**X34 0**	**X35 0**	20z3-sum(x31 to x35)	**Z3 0**
Asia	**X41 0**	**X42 0**	**X43 4**	**X44 16**	**X45 0**	20z4-sum(x41 to x45)	**Z4 1**
Africa	**X51 0**	**X52 0**	**X53 10**	**X54 0**	**X55 7**	20z5-sum(x51 to x55)	**Z5 1**
Unmet Demand	12-sum(x11 to x51)	8-sum(x12 to x51)	14-sum(x13 to x53)	16-sum(x14 to x54)	7-sum(x15 to x55)		

Using Excel Solver we find that the optimal cost is $ 23,689

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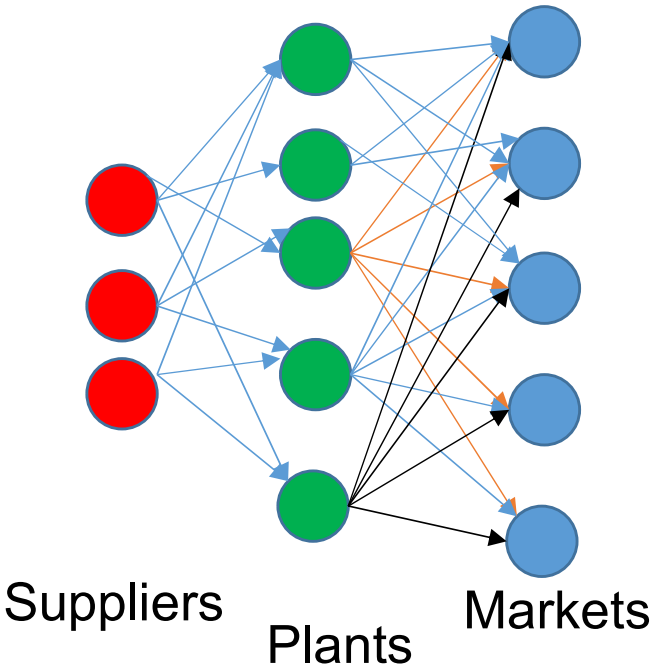
Capacitated Plant Location Problem (4) Two-Echelon Network

- Same decisions as before: which plants to open? how to allocate the markets demand?
- In addition, the logistics manager needs to select the plant suppliers and decide on raw materials quantity bought from each supplier
- Contract with the suppliers is transaction based with variable cost ($1,000 per mln liters) and annual capacity (mln) given as



	Plant Purchasing and Transp. cost per million liters					
Suppliers	N. America	S. America	Europe	Asia	Africa	Capacity
N. America	41	46	51	65	58	40
Asia	59	39	54	49	50	35
Africa	51	53	48	60	56	25

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CPLP: two-echelon

- **Decision variables**
  - Which plants to open?
  - How much to produce and transport from a plant to a region?
  - How much to buy from each supplier?
- **Decision variables**
  - z_i: 1 if plant i is open, otherwise.
  - x_{ij}: quantity to ship from i to j
  - u_{li}: quantity shipped from supplier l to plant i
- **Objective** Sum $f_i z_i$ for all i + Sum $c_{li} u_{li}$ for all l, i+ Sum $c_{ij} x_{ij}$ for all i, j

  Fixed cost Variable cost
- **Constraints**:
  - All demands should be satisfied
  - All plants and suppliers capacity limits are met
  - Output flow from a plant cannot exceed its input flow of raw material
- **Constraints**:
  - $Unmet\ demand\ D_j - \sum_{i=1}^{5} x_{ij} = 0$ and Left plant capacity $20z_i - \sum_{j=1}^{5} x_{ij} \geq 0$
  - Left supplier capacity $C_l - \sum_{i=1}^{5} u_{li} \geq 0$, balancing flows $\sum_{i=1}^{5} u_{li} \geq \sum_{j=1}^{5} x_{ij}$

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CPLP: two-echelon
Excel optimization results

Optimal cost $ 26,340

Plants	N. America	S. America	Europe	Asia	Africa	Left Capacity	Plant Open/Close	Total output Plant
N. America	12	8	0	0	0	0	1	20
S. America	0	0	0	0	0	0	0	0
Europe	0	0	0	0	0	0	0	0
Asia	0	0	4	16	0	0	1	20
Africa	0	0	10	0	7	3	1	17
Unmet Demand	0	0	0	0	0			

	Quantities from suppliers to plants					Excess
Suppliers	N. America	S. America	Europe	Asia	Africa	supplier capacity
N. America	20	0	0	0	0	20
Asia	0	0	0	20	15	0
Africa	0	0	0	0	2	23
Total quantity to plant	20	0	0	20	17	

Possible Extensions of location problems

- Network design with multi-commodity
- Uncertain demand values and uncertain cost parameters
- Including inventory (cycle + safety) cost of end products to satisfy service level constraints
- Taking into account possible disruption in plants and DCs availability
  - In this case deciding optimally on facilities backup to be used in case of unavailable

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Facility Relocation and Facility Closing (2 of 5)

- Facility relocation
  - Occurs when a firm decides that it can no longer continue operations in its present facility and must move operations to another facility to better serve suppliers or customers
  - Common reason for facility relocation
    - Lack of room for expansion at current site

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Facility Relocation and Facility Closing (3 of 5)

- Facility relocation
  - Decision involves a comparison of the advantages and disadvantages of a new site to the advantages and disadvantages of an existing location
  - Must also consider the potential consequences of relocation on their human resources (consequences that may not be easily quantified)
  - Must keep current employees informed
  - Must be prepared for unplanned glitches during transition to prevent customer service issues

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Facility Relocation and Facility Closing (4 of 5)

- Facility closing
  - Occurs when a company decides to discontinue operations at a current site because the operations may no longer be needed or can by absorbed by other facilities
  - Examples leading to facility closure include:
    - Eliminating redundant capacity in mergers and acquisitions
    - Improving supply chain efficiency
    - Poor planning
    - Insufficient volume of business

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Facility Relocation and Facility Closing (5 of 5)

- Facility closing
  - Worker Adjustment and Retraining Notification (WARN) Act
  - Human impact must be considered
    - Employees should be kept informed
    - Individuals are losing jobs and pay
    - Individuals suffer a loss of self-esteem
  - Poorly handled closings can lead to negative publicity

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Key Terms (1 of 2)

- Agglomeration (industry cluster)
- Brownfields
- Center-of-gravity approach
- Expatriate workers
- Facility closing
- Facility location
- Facility relocation
- Free trade zone
- Grid systems
- Intermodal competition
- Intramodal competition
- Inventory tax
- Locavore strategy

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Key Terms (2 of 2)

- Maquiladora plants
- Nearsourcing
- Pure materials
- Quality-of-life considerations
- Right-to-work laws
- Supplier parks
- Sweatshops
- Weight-gaining products
- Weight-losing products

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